

III. EDA-Data Cleaning

AIM:

- Handling missing values: detection, filling, and dropping
- Removing duplicates and unnecessary data
- Data type conversion and ensuring consistency
- Normalize data (e.g., standardization, min-max scaling).

```
import pandas as pd
```

```
import numpy as np
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
```

```
[1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import MinMaxScaler, StandardScaler

df = pd.read_csv("loan_data.csv")

print("Initial Data Overview:")
df.info()
```

```
Initial Data Overview:
<class 'pandas.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID               9 non-null     str
1   Gender                9 non-null     str
2   Married               9 non-null     str
3   Dependents            8 non-null     str
4   Education              9 non-null     str
5   Self_Employed         8 non-null     str
6   ApplicantIncome       9 non-null     int64
7   CoapplicantIncome     9 non-null     int64
8   LoanAmount            8 non-null     float64
9   Loan_Amount_Term      8 non-null     float64
10  Credit_History         8 non-null     float64
11  Property_Area          9 non-null     str
12  Loan_Status           9 non-null     str
dtypes: float64(3), int64(2), str(8)
memory usage: 1.0 KB
```

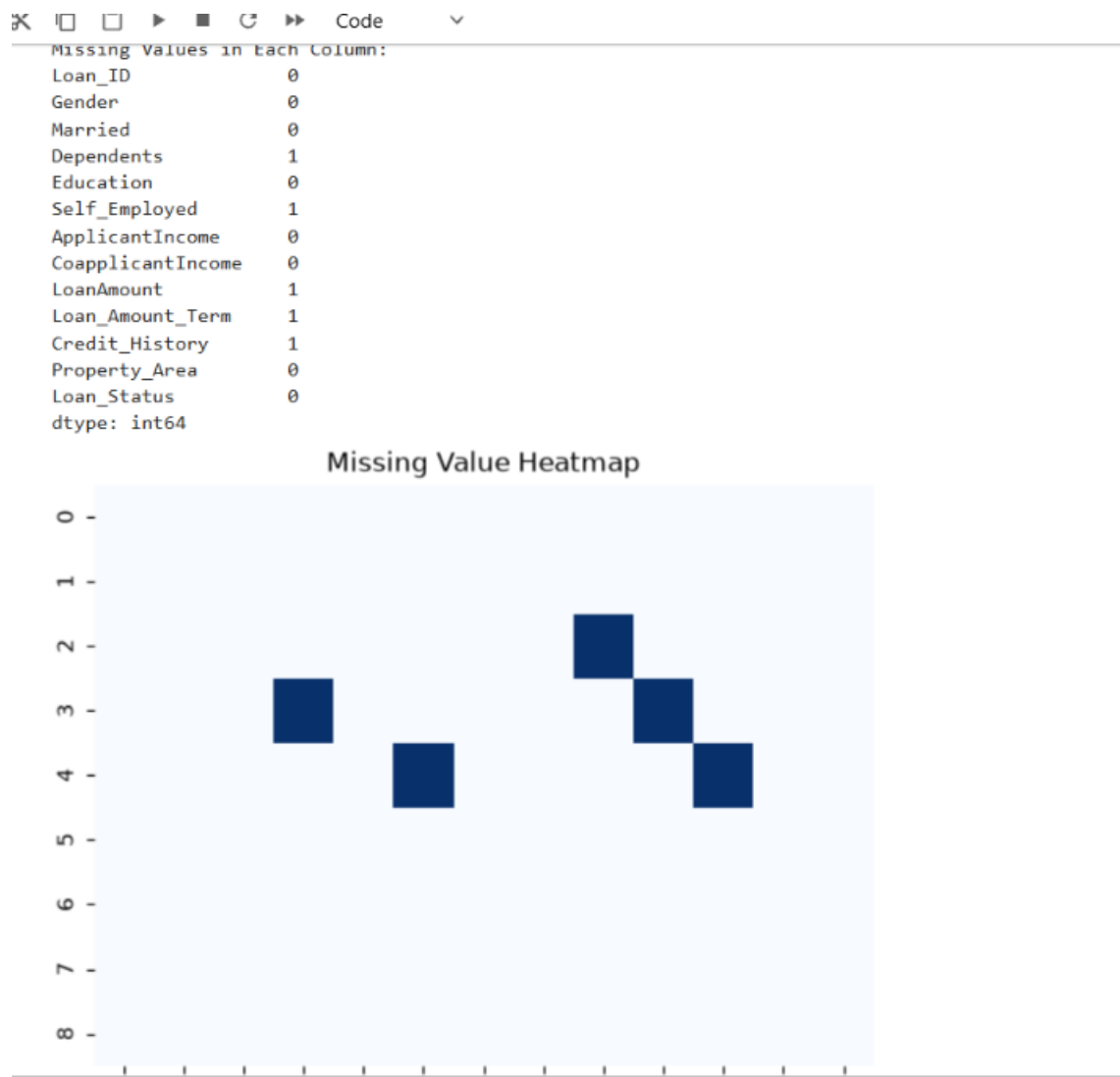
```
print("Missing Values in Each Column:")
```

```
print(df.isnull().sum())
```

```
sns.heatmap(df.isnull(), cbar=False, cmap="Blues")
```

```
plt.title("Missing Value Heatmap")
```

```
plt.show()
```



```
# Fill categorical columns with the mode
```

```
for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
```

```
    df[col] = df[col].fillna(df[col].mode()[0])
```

```
# Fill numerical columns
```

```
df['LoanAmount'] = df['LoanAmount'].fillna(df['LoanAmount'].median())
```

```
df['Loan_Amount_Term'] = df['Loan_Amount_Term'].fillna(  
    df['Loan_Amount_Term'].mode()[0]  
)
```

```
df['Credit_History'] = df['Credit_History'].fillna(  
    df['Credit_History'].mode()[0]  
)
```

```
print("Missing values after filling:")
```

```
print(df.isnull().sum())
```

```
Missing values after filling:
```

Loan_ID	0
Gender	0
Married	0
Dependents	0
Education	0
Self_Employed	0
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	0
Loan_Amount_Term	0
Credit_History	0
Property_Area	0
Loan_Status	0

dtype: int64

```
] : initial_rows = df.shape[0]  
  
df.drop_duplicates(inplace=True)  
  
print("Removed", initial_rows - df.shape[0], "duplicate rows.")
```

```
Removed 1 duplicate rows.
```

```
] : df['Dependents'] = (  
    df['Dependents']  
    .replace('3+', '3')  
    .astype(int)  
)
```

```
3      -1.732051
4       0.577350
```

```
[9]: print("Cleaned Data Summary:")

df.info()

print("\nFirst 5 rows of cleaned data:")
print(df.head())

print("\nRemaining missing values:")
print(df.isnull().sum().sum())
```

```
Cleaned Data Summary:
<class 'pandas.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                8 non-null     str
1   Gender                 8 non-null     str
2   Married                8 non-null     str
3   Dependents             8 non-null     int64
4   Education              8 non-null     str
5   Self_Employed          8 non-null     str
6   ApplicantIncome        8 non-null     float64
7   CoapplicantIncome      8 non-null     float64
8   LoanAmount             8 non-null     float64
9   Loan_Amount_Term       8 non-null     float64
10  Credit_History         8 non-null     float64
11  Property_Area          8 non-null     str
12  Loan_Status            8 non-null     str
dtypes: float64(5), int64(1), str(7)
memory usage: 964.0 bytes
```

```
First 5 rows of cleaned data:
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed \
0   L001    Male     Yes         1    Graduate         No
1   L002  Female     No         0    Graduate         No
2   L003    Male     Yes         2  Not graduate     Yes
3   L004  Female     Yes         1    Graduate         No
4   L005    Male     No         0    Graduate         No
```

```
memory usage: 964.0 bytes
```

```
First 5 rows of cleaned data:
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed \
0   L001    Male     Yes         1    Graduate         No
1   L002  Female     No         0    Graduate         No
2   L003    Male     Yes         2  Not graduate     Yes
3   L004  Female     Yes         1    Graduate         No
4   L005    Male     No         0    Graduate         No

   ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term \
0              0.4              0.500000    0.333333          360.0
1              0.2              0.000000    0.166667          360.0
2              0.6              0.666667    0.291667          360.0
3              0.1              0.400000    0.083333          360.0
4              0.3              0.000000    0.250000          360.0

   Credit_History  Property_Area  Loan_Status
0       0.577350        Urban           Y
1       0.577350        Rural           N
2       0.577350    Semiurban           Y
3      -1.732051        Urban           N
4       0.577350        Rural           Y
```

```
Remaining missing values:
0
```

RESULT:

Thus, the program was written and executed successfully.